AUTOMOBILE ENGINEERING · PART 9 OF 12

Platforms, Manufacturing and Supply Chain, and How to Read a Spec Sheet

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

A car company's biggest costs are fixed: developing the car and building the plant. So makers share a **platform** (the floor, suspension mounts and electrical architecture) across many models, so that one development bill is spread over lakhs of cars. Inside a plant, steel becomes a body, the body is painted, the powertrain is fitted and the car is inspected. Behind the plant sits a supply chain of thousands of parts suppliers, which is where disruptions (chips, rare-earth magnets, battery cells) hit. The chapter ends with a practical skill: how to read a spec sheet and turn its numbers into meaning.

1. Platforms: one design, many cars

A platform is the shared underbody and its engineering: wheelbase options, mounting points for engine and suspension, crash structure and wiring. Cars on one platform can look and feel quite different (hatchback, sedan, SUV) while sharing many unseen parts. The economics are the point: if developing a platform costs ₹2,000 crore (an illustrative number) and it underpins 5 lakh cars, development adds ₹40,000 to each car; on 1 lakh cars it adds ₹2 lakh. **Break-even volume = fixed costs ÷ (price − variable cost per car)**, so volume is what makes a car profitable.

Platform	Maker	Cars built on it (examples)
Heartect	Maruti Suzuki	Swift, Dzire
ALFA-ARC	Tata Motors	Punch, Altroz (petrol)
acti.ev	Tata Motors	Punch.ev (first car on it)
INGLO	Mahindra	BE 6, XEV 9e (electric, rear-motor layout)
MQB-A0-IN	Volkswagen Group (Skoda, VW)	Kushaq, Slavia, Taigun, Virtus
TNGA GA-C*	Toyota	Innova Hycross

Notice the electric platforms: Tata and Mahindra built dedicated EV platforms (acti.ev, INGLO) rather than modifying petrol cars. A purpose-built floor lets the battery sit flat under the cabin, which improves space and handling (Part 6).

2. Inside a car plant

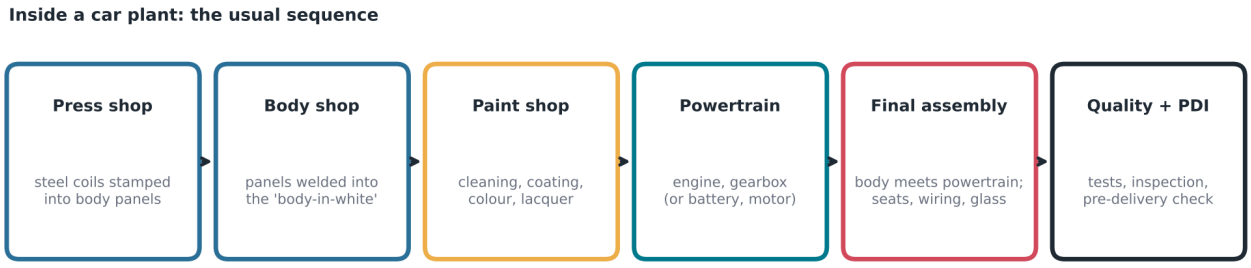


Fig 1. A typical assembly sequence. The 'body-in-white' is the bare welded shell before paint and parts.

- **Press shop:** giant presses stamp steel coils into doors, roofs and floors.

- **Body shop:** robots weld hundreds of panels into the body-in-white. Weld quality decides crash strength.
- **Paint shop:** the body is cleaned, rust-proofed and coated; this is the most energy- and water-intensive stage.
- **Powertrain:** engines and gearboxes are made or assembled (or, for EVs, battery packs and motors), usually in a separate line or plant.
- **Final assembly:** the body and powertrain are 'married'; seats, wiring, glass and wheels are fitted on a moving line.
- **Quality and PDI:** tests, inspection and a pre-delivery inspection before the car goes to a dealer.

Major Indian plants include Maruti Suzuki's at Manesar and Gurugram, Tata's at Sanand, Mahindra's at Chakan and Hyundai's near Chennai.* Plant capacity, shifts and how fast a model can be sold set a car's waiting period.

3. The supply chain

A car has around 30,000 parts.* The maker designs, assembles and brands the car, but buys most parts from suppliers. **Tier 1** suppliers (seats, electronics, transmissions) deliver complete systems to the plant; **Tier 2** and **Tier 3** supply components and raw materials to them. A shortage anywhere stops the line.

- **Localisation:** the share of parts made in India. Higher localisation cuts import duty, currency risk and cost, and is a stated goal of programmes such as the government's production-linked incentive (PLI) scheme for the auto sector (approved in September 2021 with an outlay of about ₹26,000 crore).
- **Semiconductors:** the chip shortage of 2021–22 forced several makers to cut production and extended waiting periods.*
- **EV-specific risks:** battery cells and rare-earth magnets (used in many EV motors) are largely imported, so EV costs and supply are exposed to overseas policy and prices.*
- **Just-in-time:** many suppliers deliver hours before parts are needed, which keeps stock low but makes disruptions costly.

4. How to read a spec sheet

A spec sheet is a list of numbers; the skill is to ask what each one tells you. Use the Swift's figures from earlier parts as an example.

Number	What it tells you	Swift example and what to watch
Engine cc, cylinders, aspiration	Size and style of the engine (Part 1)	1,197 cc, 3 cylinders, natural. Compare power per litre, not cc.
Power (bhp @ rpm)	Top-end performance (Part 2)	80.46 bhp @ 5,700 rpm: you must rev to use it.
Torque (Nm @ rpm)	Pulling strength and its spread	111.7 Nm @ 4,300 rpm: a narrow, higher-rpm peak.
Kerb weight	Power-to-weight, efficiency	About 925 kg (listing)*, so about 87 bhp per tonne.
Transmission	How it drives (Part 3)	5-speed manual or 5-speed AMT.
ARAI mileage	Comparison tool, not a promise (Part 2)	Expect real figures well below the claim.
Fuel tank	Range = tank × real km/l	37 litres (listing).
Boot space	Luggage practicality	265 litres (listing).

Number	What it tells you	Swift example and what to watch
Ground clearance, turning radius	Rough-road ability, city manoeuvres (Part 4)	Check both on the brochure.
Airbags, ESC, NCAP score	Safety (Part 5)	Variant matters; check the tested version.

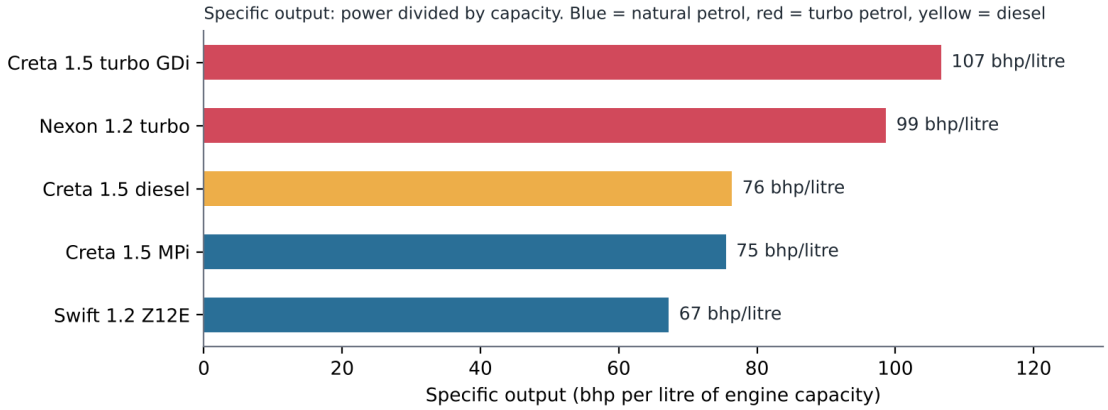


Fig 2. Specific output, power per litre of capacity, for engines in Part 1. Boosted engines extract 40–50% more power per litre than natural petrols.

Two derived numbers are worth computing for any car. **Specific output** (power ÷ litres) compares engines of different sizes: the Swift manages 67 bhp per litre, the Nexon 1.2 turbo 99 and the Creta turbo 107. **Power-to-weight** (power ÷ tonnes) compares how lively cars feel: the Swift's 87 bhp per tonne is a good benchmark for a small, light car.* Together with torque spread, these two numbers explain most of what you will feel on a test drive.

Summary

- A platform is the shared underbody; reusing it spreads fixed development costs across many cars.
- Break-even volume = fixed costs ÷ (price − variable cost); volume decides profit.
- A plant runs press shop, body shop, paint shop, powertrain, final assembly and quality checks.
- Supply chains run in tiers; chips, rare-earth magnets and battery cells are the big risks.
- Read a spec sheet by asking what each number implies; compute specific output and power-to-weight.
- Always compare the exact variant: airbags, gearbox and weight differ by trim.

*Unverified or general knowledge: the TNGA GA-C platform name for the Hycross, ALFA-ARC and acti.ev details, the plant locations, the number of parts in a car, the 2021–22 semiconductor shortage and EV-component import exposure, and the Swift's kerb weight (from a listing) were not checked against official documents. The ₹2,000 crore platform cost and volumes are made-up numbers to show the arithmetic. Cross-checked in November 2026: INGLO (BE 6, rear motor), MQB-A0-IN (Kushaq, Slavia, Taigun, Virtus) and Heartect (Dzire). Spec-sheet figures: Wikipedia and Autocar India listings. The auto PLI scheme (approved September 2021, about ₹26,000 crore; Business Standard) was cross-checked in November 2026.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 10.